# 1. Lab 07: Router Architecture & Forwarding

**Lab Name:** Lab 07: Router Architecture & Forwarding
**Date:** YYYY-MM-DD
**Name:** [Your Name]
**Lab Partners:** [Partners' Names, if any]
**Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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## AI Policy & Acceptable Use Guidelines

 AI Policy & Acceptable Use Guidelines - Yellow Icon

Artificial Intelligence tools can be powerful learning aids, but they must be used ethically and responsibly to ensure you develop core networking skills.

- **Lab Assignments:** Do not use AI to perform lab steps or write your lab submissions. However, if you get stuck on a concept or technical obstacle, you may use AI as a virtual tutor to ask clarifying questions and help you troubleshoot.

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# 2. Background

Routers are Network Layer devices responsible for forwarding packets end-to-end between source and destination hosts. A router's internal architecture is logically split into two cooperating planes:

1. **Control Plane (Software / Routing Information Base - RIB)**:
    - Managed by the router CPU in the millisecond timeframe.
    - Runs routing protocols (e.g., OSPF, BGP), processes static routes, and maintains the master **Routing Information Base (RIB)**.
    - The RIB contains the complete logical topology and recursive route lookups (e.g., "*To reach prefix 10.1.23.0/24, send to next-hop IP 10.1.12.2*").
2. **Data Plane (Hardware / Forwarding Information Base - FIB & Adjacencies)**:
    - Operates in the nanosecond timeframe directly on line cards and interface silicon.
    - **How the FIB is Calculated**:
        - The control plane compiles and pre-processes the RIB into the **Forwarding Information Base (FIB)**.
        - Unlike the RIB, the FIB resolves all recursive next-hops in advance. Every destination prefix in the FIB points directly to an egress physical interface.
        - The FIB is structured as an optimized lookup tree (mtrie) or programmed into **Ternary Content Addressable Memory (TCAM)** hardware, allowing the router to perform **Longest Prefix Matching (LPM)** in a single clock cycle at line rate.
    - **The Adjacency Table (Layer 2 Rewrite)**:
        - In parallel, the router derives an **Adjacency Table** from ARP and neighbor discovery caches.
        - For each next-hop IP, the Adjacency Table pre-computes the complete Layer 2 Ethernet frame header (Destination MAC, Source MAC, and EtherType).
    - **The Forwarding Cycle (Match-Plus-Action)**: When an ingress packet arrives:
        1. **Match**: The destination IP is matched against the FIB in hardware to find the longest matching prefix.

2. **Action**: The FIB pointer references the Adjacency Table to immediately overwrite the Layer 2 frame header, decrement the IPv4 TTL, update the checksum, and push the frame across the switching fabric to the designated output port without CPU interruption.

In this lab, you will configure a linear chain of Cisco IOSv routers, verify static routing and transit forwarding, and query the router data-plane tables (Cisco Express Forwarding / CEF and Adjacency) to see how theoretical router architectures are implemented in real networking devices.

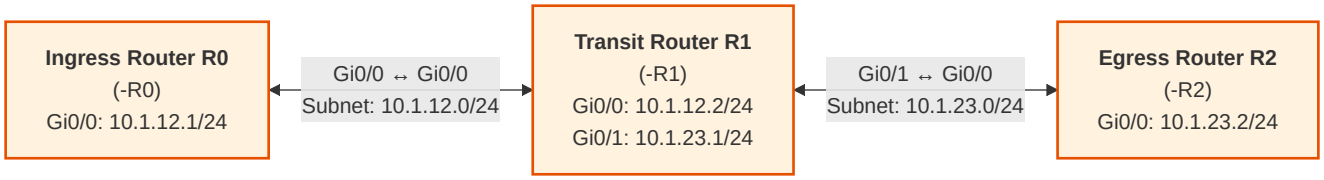
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# 3. Objective / Purpose

To configure static routing across a multi-router transit network, analyze the difference between the routing table (RIB) and the forwarding table (FIB), inspect Layer 2 rewrite adjacency tables (Cisco Express Forwarding), and map theoretical router architecture components to CLI commands and packet captures.

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# 4. Topology Diagram




**Details:**

- 3x Cisco IOSv Routers connected in a linear transit chain ( `R0 <-> R1 <-> R2` ).
- Node names: `<initials>-R0` (Ingress), `<initials>-R1` (Transit Core), `<initials>-R2` (Egress).

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# 5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 3x Cisco IOSv Routers

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# 6. Configuration / Methodology

### IP Addressing & Interface Connection Table

| Device | CML Node Name | Interface | IP Address | Subnet Mask | Connected To Interface | Subnet Network |
|---|---|---|---|---|---|---|
| **Ingress Router** | `<initials>-R0` | `Gi0/0` | `10.1.12.1` | `255.255.255.0` (`/24`) | `<initials>-R1 Gi0/0` | `10.1.12.0/24` (Subnet 1) |
| **Transit Router** | `<initials>-R1` | `Gi0/0` | `10.1.12.2` | `255.255.255.0` (`/24`) | `<initials>-R0 Gi0/0` | `10.1.12.0/24` (Subnet 1) |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Transit Router** | `<initials>-R1` | `Gi0/1` | `10.1.23.1` | `255.255.255.0` (`/24`) | `<initials>-R2 Gi0/0` | `10.1.23.0/24` (Subnet 2) |
| **Egress Router** | `<initials>-R2` | `Gi0/0` | `10.1.23.2` | `255.255.255.0` (`/24`) | `<initials>-R1 Gi0/1` | `10.1.23.0/24` (Subnet 2) |

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## Step-by-Step Configuration Guide

### 1. Configure Ingress Router R0 ( `<initials>-R0` )

Open R0's console, elevate privileges, and configure IP addressing and static routing toward Subnet 2:

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R0
KH-R0(config)# interface GigabitEthernet0/0
KH-R0(config-if)# ip address 10.1.12.1 255.255.255.0
KH-R0(config-if)# no shutdown
KH-R0(config-if)# exit
KH-R0(config)# ip route 10.1.23.0 255.255.255.0 10.1.12.2
KH-R0(config)# end
KH-R0# write memory
```

*(The static route directs traffic destined for Subnet 2 `10.1.23.0/24` to next-hop `10.1.12.2` on transit router R1).*

### 2. Configure Transit Core Router R1 ( `<initials>-R1` )

Open R1's console and configure both interfaces:

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R1
KH-R1(config)# interface GigabitEthernet0/0
KH-R1(config-if)# ip address 10.1.12.2 255.255.255.0
KH-R1(config-if)# no shutdown
KH-R1(config-if)# exit
KH-R1(config)# interface GigabitEthernet0/1
KH-R1(config-if)# ip address 10.1.23.1 255.255.255.0
KH-R1(config-if)# no shutdown
KH-R1(config-if)# end
KH-R1# write memory
```

*(Note: Because R1 is directly connected to both `10.1.12.0/24` and `10.1.23.0/24`, its routing table automatically knows how to forward packets between both subnets without any static routes!).*

### 3. Configure Egress Router R2 ( `<initials>-R2` )

Open R2's console and configure its interface and return static route toward Subnet 1:

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R2
```

```
KH-R2(config)# interface GigabitEthernet0/0
KH-R2(config-if)# ip address 10.1.23.2 255.255.255.0
KH-R2(config-if)# no shutdown
KH-R2(config-if)# exit
KH-R2(config)# ip route 10.1.12.0 255.255.255.0 10.1.23.1
KH-R2(config)# end
KH-R2# write memory
```

*(The return static route directs packets destined for Subnet 1 `10.1.12.0/24` to next-hop `10.1.23.1` on transit router R1).*

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### Step-by-Step Traffic Generation & Execution:

> *[!IMPORTANT]* ***REMINDER: All screenshots, console outputs, and packet captures (.pcap files) must be taken from the SAME active lab session. Mixing screenshots or trace files across different runs or sessions is not permitted.***

1. **Verify End-to-End Connectivity**: Open `<initials>-R0` 's console and ping across transit router R1 to reach R2:

   ```
   KH-R0# ping 10.1.23.2
   ```

   *(Ensure you get a success rate of 100 percent `!!!!!` , verifying that transit routing is fully operational).*

2. **Start Packet Capture**: Right-click the link between **Router R0 and Router R1**, select **Packet Capture**, set filter to `icmp or arp` , and click **Start**.

3. **Trigger ARP Resolution and ICMP Forwarding**: Open `<initials>-R0` 's console and clear the ARP cache to force a fresh address resolution:

   ```
   KH-R0# clear arp
   KH-R0# ping 10.1.23.2
   ```

4. **Collect Router Data Plane CLI Outputs**: Run the following verification commands on `<initials>-R0` and record the results:

   - `show ip route` (Displays the control plane Routing Information Base / RIB).
   - `show ip cef` (Displays the data plane Forwarding Information Base / FIB).
   - `show adjacency detail` (Displays the Layer 2 rewrite string and hardware adjacency table).
   - `show arp` (Displays the resolved IP-to-MAC associations).

5. **Collect Transit Router CLI Output on `<initials>-R1`** :

   - `show ip route` (Notice how both subnets appear as directly connected `C` ).
   - `show ip cef` (Notice how entries for both subnets are marked as attached).

6. **Download Capture**: Stop the CML packet capture and download the `.pcap` capture file to your local computer.

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# 7. Verification and Analysis Questions

Analyze the CLI outputs collected from your routers and the packet capture in Wireshark, and answer the following questions:

### 1. Control Plane vs. Data Plane CLI Mapping

- Examine the outputs of `show ip route` and `show ip cef` on `<initials>-R0`.
- Contrast the two outputs. Which command displays the RIB (Routing Information Base) belonging to the **control plane**, and which displays the FIB (Forwarding Information Base) belonging to the **data plane**?
- Explain the difference in operation speed (millisecond vs. nanosecond timeframe) and hardware implementation between these two planes.

### 2. Longest Prefix Matching

- In destination-based forwarding, how does a router determine which table entry to use when a packet destination IP matches multiple route entries?
- Does Cisco IOS use longest prefix matching? Illustrate how the IP prefix `10.1.12.0/24` in your routing table corresponds to a range of addresses, and explain what entry is matched when a packet arrives destined for `10.1.23.2`.

### 3. CEF Adjacency and Layer 2 Rewrite (Match-plus-Action)

- Inspect the output of the `show adjacency detail` command on `<initials>-R0`.
- Look at the hexadecimal output representing the Layer 2 rewrite string (adjacencies). Explain how this maps to the "link layer encapsulation" function performed in a router's **output port** interface card before sending bits onto the physical link.
- When R0 forwards a packet destined for `10.1.23.2`, whose MAC address is placed in the destination MAC field of the Ethernet frame sent out `Gi0/0`? (Hint: Is it R2's MAC or next-hop R1's MAC?).
- In what way does this represent a "Match-plus-Action" model?

### 4. Router Switching Fabric Architecture

- What are the three primary types of **switching fabrics** traditionally used inside routers (memory, bus, and crossbar/interconnection network)?
- Cisco IOSv is a software-based virtual router running inside a hypervisor on x86 server hardware. Which switching fabric architecture does it functionally simulate/rely on for transferring packets between virtual interfaces? What is the main bottleneck of this fabric type?

### 5. Queuing and HOL Blocking

- Define **Head-of-the-Line (HOL) blocking** and explain under what circumstances it occurs (does it happen at the input ports or output ports of a switching fabric?).
- If a router's switching fabric is slower than the combined input ports, where will queuing occur? If the fabric is faster, where will queuing occur?

### 6. Buffer Management and Sizing

- According to the standard rule of thumb (RFC 3439), how much buffer capacity is recommended for a router interface operating at speed $C$ with a round-trip time $RTT$?
- If one of `<initials>-R0` 's interfaces was a physical 10 Gbps port with a typical network RTT of 120 ms, calculate the required buffer size in Megabits.
- Explain the negative consequence of having too much buffering on a link (bufferbloat).

### 7. Scheduling and Net Neutrality

- Briefly define the **Weighted Fair Queueing (WFQ)** scheduling policy. How does it guarantee bandwidth to specific traffic classes compared to simple FCFS (FIFO)?
- Explain how packet scheduling and buffer management are technical mechanisms directly related to the social, political, and economic arguments surrounding **Network Neutrality** (specifically the rules against throttling

and paid prioritization).

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# 8. Troubleshooting

**Issue Encountered:** [Describe what went wrong, if anything]
**Discovery Method:** [How did you find the issue?]
**Resolution:** [How did you fix it?]

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# 9. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- The Wireshark packet capture file ( `r0_r1_traffic.pcap` or `r0_r1_traffic.pcapng` ) containing the captured ARP and ICMP traffic between R0 and R1.
- A single PDF report containing:
  - Your written answers to the 7 analysis questions in Section 7.
  - A screenshot of your active CML topology showing the running linear chain of routers ( `<initials>-R0` , `<initials>-R1` , `<initials>-R2` ).
  - A screenshot of the routing table ( `show ip route` command output) for **each** of the three routers ( `<initials>-R0` , `<initials>-R1` , and `<initials>-R2` ).
  - The required screenshot of the router CLI console displaying `<initials>-R0` 's `show ip interface brief` status and `show ip cef` output.
  - An annotated screenshot of your Wireshark capture window showing the ARP Request and ARP Reply exchange between R0 and R1 along with the ICMP Echo Request/Reply packets.